

KUSHAGRA TYAGI

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TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, HTML5, CSS3

Backend: Spring Boot, Spring MVC, Spring Security, Hibernate, RESTful APIs

Frontend: React.js, Zustand, Bootstrap, Tailwind CSS

Database: MySQL, Redis

Tools: Docker, Git, GitHub, Maven, IntelliJ IDEA

PROJECTS

TradeVerse – Simulated Indian Stock Market & Trading Platform |

Jan 2026 - Present

Spring Boot, React.js, MySQL, WebSocket/STOMP, Redis

A full-stack replicated Indian stock market platform with a real-time order matching engine, 200 trading bots, a news/sentiment engine, strategy backtester, and cognitive bias analytics for trading education

- Engineered a high-throughput simulated stock exchange order matching engine, processing over 1,000 bid/ask queue updates per second with zero-latency UI streaming.
- Launched 200 automated trading bots across 4 behavioral types (momentum, market-maker, sentiment, whale) that continuously generate order flow, driving realistic price discovery and market depth.
- Developed a real-time sentiment analysis engine that models macro and micro market conditions, enabling scenario-based learning for users.
- Designed a strategy backtester and historical scenario simulator allowing users to replay past market conditions and test custom trading rules against stored tick data.
- Built a trade journal with cognitive bias analytics that identifies patterns (loss-aversion, overtrading, revenge trading) from a user's trade history using rule based scoring
- Architected as a modular monolith with clear domain boundaries (market, trading, analytics, user) with Redis used for orderbook caching and session management

Shopnest- Full-Stack E-Commerce platform |

Sept 2025 - Dec 2025

Spring Boot, React.js, MySQL, Vercel

A full-stack e-commerce platform with a dynamic pricing engine, group buying discounts, and bill splitting built on React, Spring Boot, and MySQL.

- Launched a complete e-commerce application with product catalog cart, checkout, and order management: React frontend deployed on Vercel, Spring Boot backend is on cloud VM
- Implemented a dynamic pricing engine that adjusts product prices based on inventory levels, demand signals, and time-based rules.
- Built group buying and bill-splitting features enabling users to pool purchases for tiered discounts, with real-time participant tracking and automatic price recalculation.
- Designed a normalized MySQL schema with optimized indexing; used Spring Data JPA with custom JPQL queries for filtering and pagination.

ACHIEVEMENTS

- Solved 300+ problems on LeetCode across arrays, linked lists, trees, graphs, DP; following Striver's A2Z DSA sheet
- Submitted a proposal to GSOC 2026 under the C2SI organization for the CodeLabz project, a platform for creating and sharing interactive coding labs

EDUCATION

B.Tech in Computer Science

2024-2028

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY , Greater Noida | AKTU | CGPA:7.7/10